Assignment - 1

Code-

Recursive

```
#include <iostream>
using namespace std;

int fibonacci(int n) {
    if (n <= 1)
        return n;

    return fibonacci(n - 1) + fibonacci(n - 2);
}

int main() {
    int n;

    cout << "Enter the number of terms: ";
    cin >> n;

    if (n <= 0) {
        cout << "Enter a positive number.";
        return 0;
    }

    cout << "Fibonacci Series: ";
    for (int i = 0; i < n; i++)
        cout << fibonacci(i) << " ";

    cout << "\nTime Complexity: O(2^n)";
    cout << "\nSpace Complexity: O(n)";

    return 0;
}
```